

Time Series, Clustering, Recommenders, Anomaly Detection

Forecasting, segmentation, recommendation and outlier detection: methods, parameters, metrics and the traps in each.

DS Prep
INTERVIEW CHEAT
SHEET

TIME SERIES BASICS

- Components: **trend**, **seasonality** (fixed period), **cycle**, **noise**. Additive: $y = T + S + e$; multiplicative: $y = T \times S \times e$ (seasonal swing grows with level; take logs).
- **Stationary**: constant mean, variance and autocovariance. Make stationary by differencing ($y_t - y_{t-1}$), seasonal differencing (lag m), log transform. Test with **ADF** (null: non-stationary) / **KPSS**.
- **ACF/PACF** plots: ACF cuts off at $q \rightarrow$ MA(q); PACF cuts off at $p \rightarrow$ AR(p). Autocorrelation at lag 7 = weekly pattern.
- **Always beat a baseline**: naive (last value), seasonal naive (same weekday last week), moving average.

SMOOTHING FORMULAS AND WORKFLOW

- **Simple exponential smoothing**: $\text{level}_t = \alpha \cdot y_t + (1 - \alpha) \cdot \text{level}_{t-1}$; α near 1 reacts fast, near 0 is smooth. Holt adds a trend term, Holt-Winters a seasonal term.
- Moving average of window w : lags the series by about $w/2$.
- Workflow: plot $\rightarrow$ decompose (STL) $\rightarrow$ baseline $\rightarrow$ stationarity check $\rightarrow$ model $\rightarrow$ **residuals should look like white noise** (Ljung-Box test, no ACF spikes) $\rightarrow$ forecast with intervals $\rightarrow$ backtest.
- Missing timestamps: reindex to a regular calendar, fill (interpolate for sensors, 0 for counts), flag outliers, mind time zones and daylight saving.
- Many series: global model (one GBM across series with series-id features) often beats one ARIMA per series. Intermittent demand (many zeros): Croston, or count models.

FORECASTING MODELS

Model	Idea	Use
Moving average / exponential smoothing	weighted recent history	short horizon, level only
Holt-Winters (ETS)	level + trend + season	seasonal business series
ARIMA(p, d, q)	AR(p) past values, d differences, MA(q) past errors	single stationary-ish series
SARIMA	ARIMA + seasonal (P, D, Q, m)	seasonality
SARIMAX / regression	add external regressors	promotions, price, weather
Prophet	trend + seasonality + holidays	business series with gaps, holidays
GBM / LightGBM on lag features	lags, rolling stats, calendar	many series, rich features
LSTM / Transformer	sequence nets	long histories, many series

- Feature recipe: lags (1, 7, 28), rolling mean/std, expanding stats, differences, day of week, month, holiday, price/promo. Use only information known at forecast time.

VALIDATION AND METRICS

- **Never random-split**. Use a temporal holdout and **rolling-origin / forward-chaining** CV (`TimeSeriesSplit`); gap between train and test if features look ahead.
- Metrics: **MAE**, **RMSE**, **MAPE** (breaks near 0), **sMAPE**, **MASE** (vs naive; < 1 beats naive). Evaluate by horizon; add prediction intervals.

- Pitfalls: leakage from future data (rolling windows computed centred), ignoring holidays/structural breaks, using ARIMA on non-stationary data, forecasting too far ahead, tuning on one split.

```
from statsmodels.tsa.arima.model import ARIMA
m = ARIMA(y, order=(1, 1, 1)).fit()
m.forecast(steps=7)
from statsmodels.tsa.stattools import adfuller
adfuller(y)[1] # p < 0.05: stationary
from sklearn.model_selection import TimeSeriesSplit
```

CLUSTERING

- Unsupervised grouping; **scale features first**; no ground truth, so judge by metrics and usefulness.

Method	Idea and limits
K-means	assign to nearest centre, recompute means, repeat. Fast; needs k , assumes round similar-size clusters, sensitive to scale, outliers and init (use <code>k-means++</code>)
Hierarchical	merge closest groups, cut the dendrogram. No k up front; $O(n^2)$; linkage choice (Ward, single, complete)
DBSCAN	dense regions via <code>eps</code> , <code>min_samples</code> ; noise = -1 . Arbitrary shapes, finds outliers; struggles with varying density
GMM	soft assignments with Gaussians (EM). Elliptical clusters, probabilities; needs k

- **Choose k** : elbow of inertia (WCSS), **silhouette** score (-1 to 1), gap statistic, stability across seeds, business sense.
- Mixed data: Gower distance / k -prototypes. High dimensions: PCA first. Evaluate with labels (ARI, NMI) only if available.
- Profile clusters (means per feature, sizes), name them, test they are stable and actionable. K-means distance is Euclidean: standardise.

RECOMMENDER SYSTEMS

- **Content-based:** recommend items similar to what the user liked (item features, TF-IDF, embeddings, cosine similarity). No cold start for items; narrow, filter-bubble risk.
- **Collaborative filtering:** use behaviour. **User-user / item-item** neighbourhoods (cosine, Pearson); **matrix factorisation** (ALS, SVD): $R \sim U \text{ times } V\text{-transpose}$, learn k-dim vectors so the dot product predicts the rating; implicit feedback (clicks, views) needs confidence weighting.
- **Hybrid and deep** models (two-tower, neural CF, sequence models) combine signals.
- **Two stages at scale:** candidate generation (retrieve hundreds from millions with approximate nearest neighbours) then a **ranking** model (GBM / NN using user, item, context features), then re-ranking (diversity, business rules).
- **Cold start:** new user: popularity, context (location, device), onboarding questions; new item: content features, exploration.
- **Metrics:** offline precision@K, recall@K, MAP, **NDCG@K**, hit rate, coverage, diversity, novelty; RMSE only for explicit ratings. Online: A/B on CTR, conversion, retention, revenue.
- **Pitfalls:** popularity bias, feedback loops, position bias in clicks, offline-online mismatch, sparsity, leakage (use time-based splits), implicit data has no negatives (sample them).

ANOMALY DETECTION

- Define "anomalous" for the business first: point (one odd value), contextual (odd for the time/segment), collective (odd pattern).

Method	Idea	Note
z-score	$\text{abs}(x - \text{mean}) / \text{SD} > 3$	assumes normal; outliers inflate mean and SD
IQR rule	outside Q1 – 1.5 IQR, Q3 + 1.5 IQR	robust, simple
Median/MAD	robust $z = 0.6745 (x - \text{median}) / \text{MAD}$	skewed data
Isolation Forest	random splits isolate outliers in few steps	high dimensions, fast
LOF	density versus neighbours	local anomalies
One-class SVM	boundary around normal data	small data, scale needed
Autoencoder	high reconstruction error = odd	complex data, NN
Time series	residuals from a forecast, seasonal decomposition (STL)	check against the pattern

- Example: orders 98-103 normal, one day 180: mean 108.9, SD 26.7 $\rightarrow z = 2.7$ (missed); IQR fence 108.5 $\rightarrow$ flagged.
- Evaluation with few labels: precision@K by expert review, injected synthetic anomalies, alert fatigue (false alarms per day). Handle drift: what is normal changes; retrain, use rolling baselines.
- Supervised fraud: imbalance playbook (Module 8) + rules + model.

```
from sklearn.ensemble import
IsolationForest
m = IsolationForest(contamination=0.01,
random_state=0).fit(X)
flag = m.predict(X) == -1      #
anomalies
score = -m.score_samples(X)   # higher
= odder
```

SIMILARITY AND ASSOCIATION RULES

Measure	Use
Cosine similarity	direction of vectors (text, ratings); ignores magnitude
Pearson correlation	rating similarity after centring users
Jaccard	overlap of sets (items bought)
Euclidean	dense numeric features (scale first)

- **Market basket / association rules:** support = $P(A \text{ and } B)$; confidence = $P(B | A)$; **lift** = confidence/ $P(B)$ (> 1 means positive association). Apriori / FP-growth. Beware: popular items give high confidence but lift ~ 1 .
- Evaluate recommenders offline with a **time-based split** (train on the past, test on later interactions), then confirm with an A/B test: offline gains often vanish online.
- **Exploration:** bandits (epsilon-greedy, Thompson) to learn about new items; avoid showing only what is already popular.

TIME SERIES ANOMALIES AND MONITORING

- Forecast, then flag residuals beyond ~ 3 SD or beyond the prediction interval. Use STL (seasonal-trend-loess) to remove trend and seasonality, then apply z/IQR rules to the remainder.
- Alerting: tune the threshold for false alarms per day; use rolling windows; compare with the same weekday last week.
- Drift in what is "normal" (new product launch, holiday) causes alert storms: allow manual overrides and retrain.

METHOD CHOOSER

Task	Start with
Forecast a daily metric	seasonal naive $\rightarrow$ Holt-Winters/Prophet $\rightarrow$ GBM with lags
Customer segments	K-means on scaled RFM features; check silhouette
Odd-shaped clusters / noise	DBSCAN
"Customers like you also bought"	item-item CF, then factorisation
New users, little data	popularity + content
Detect fraud with few labels	rules + Isolation Forest, review top-K

SAY IT IN 20 SECONDS

- "Time series: respect order, beat a naive baseline, validate forward in time."
- "K-means needs scaled features and a chosen k; DBSCAN finds odd shapes and noise."
- "Recommenders retrieve, then rank; cold start needs content and popularity."
- "A z-score can miss an outlier because the outlier inflates the SD."